

Ethan Barr

Founding Engineer • Backend Systems, Automation & Product Support
Baltimore, MD • Open to Remote/Hybrid

SUMMARY

Founding engineer (employee #4) with 4+ years building Django/DRF backend systems, full-stack product workflows, and operational tooling at a fast-growing mining infrastructure startup. Now owns the engineering side of customer support end to end, from escalation and **root-cause analysis** through fix, deploy verification, and the response back to the customer, after building the staging simulation and trigger-testing tooling that made those issues reproducible in the first place.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, SQL, HTML, CSS

Backend: Django, Django REST Framework, FastAPI, Node.js, REST APIs, MySQL, Redis, Data Modeling

Frontend: React, Next.js, Tailwind CSS, Reusable Components, Responsive UI

Cloud & Data: Google Cloud Platform, BigQuery, Data Exports, Operational Reporting, Production Debugging

DevOps & Infra: Docker, Linux, systemd, Tailscale, Prometheus, Grafana, GitHub Actions

Support & Reliability: Incident Triage, Root Cause Analysis, Observability, Staging Simulation, Jira, Zendesk

AI & Automation: GitHub Apps, LLM Tooling, Prompt Design, Workflow Automation, AI-assisted Development

EXPERIENCE

Product Support Engineer — Foreman

August 2026 — Present

Own the engineering side of the support process, from escalation through root cause, fix, and verification

Remote / Baltimore, MD

- **Defined and proposed the Product Support Engineer** role at the company, including ownership boundaries, prioritization tiers, escalation criteria, and success measures, then took ownership of the engineering support queue.
- Own escalated customer issues end to end: reproduction, **root-cause analysis across API, database, cache, and cloud function layers**, the code fix, regression tests, PR review, deploy verification, and the technical summary back to Customer Success.
- Serve as the **single engineering point of contact for Customer Support**, absorbing unplanned support interruptions on behalf of the engineering team and escalating to domain owners when specialized ownership is required.
- Convert recurring support issues into **permanent fixes** through code changes, improved logging and observability, and reusable internal investigation documentation.

Software Engineer — Foreman

2023 — August 2026

Founding engineer (employee #4). Backend, frontend, automation, data workflows, and production operations for mining infrastructure software

Remote / Baltimore, MD

Backend & data systems

- Built and maintained **Django/DRF APIs** supporting miner inventory, issue workflows, tagging, exports, power controls, and operational workflows across large asset datasets.
- Improved reliability and performance through **query optimization, caching, data-model cleanup, and production debugging** of customer-impacting workflows.

Automation & operational tooling

- Implemented **trigger-driven automation** for actions such as reboots, tagging, and operational state changes, using Redis-backed coordination and guardrails to prevent unsafe repeated actions.
- Built internal workflows that helped support and operations teams move from manual investigation toward **repeatable tools, exports, and guided actions**.

Testing & staging infrastructure

- Built a **trigger testing harness** that overrides production wait timers and safety blockers in staging, making it possible to continuously exercise new trigger types that otherwise could not be tested without waiting on real-world timing conditions.
- Created a **staging fleet simulator** that mass-produces synthetic miners across varied issue states and streams simulated live-looking telemetry into them, so operational workflows and customer-reported conditions can be reproduced against realistic fleet data before release.

Frontend & product workflows

- Developed and refined **React / Next.js** interfaces for operational dashboards, issue management, exports, and customer-facing workflows.
- Translated support feedback and ambiguous product requests into practical UI/API changes, balancing usability, correctness, and rollout risk.

Startup ownership & cross-functional execution

- Helped scale engineering practices from the founding team onward, working directly with support, product, and stakeholders on production issues and feature delivery.
- Served in **customer-facing support rotations and early product demos**, partnering with Customer Success and customers to triage issues, communicate timelines, and turn workflow feedback into timely engineering fixes.
- Reviewed implementation approaches, clarified technical tradeoffs, and shipped fixes where backend behavior, frontend state, and operational expectations needed to line up.

Software Engineering Intern — National Committee for Quality Assurance (NCQA)

May 2020 — August 2021

Remote

- Built and tested **React/TypeScript components** for internal file review workflows, including parsing file types and displaying relevant review information.
- Collaborated on debugger and validation tooling for medical language logic, improving how internal users inspected and reviewed complex rule behavior.
- Created test environments, mocks, and unit tests to validate application behavior and reduce regressions during development.

PROJECTS

ChangeLogScribe

GitHub Apps, Stripe, email automation, SaaS workflows •

- Built a SaaS product that turns **GitHub activity into structured changelogs and release notes** for engineering teams.
- Implemented **GitHub App integration, Stripe billing, and email notifications** with an emphasis on reliable automation and a clean product experience.
- Designed the workflow around a practical developer path: connect a repository, generate release-ready summaries, and reduce manual changelog writing.

Homelab Operations Dashboard

Next.js, FastAPI, Docker, Linux, systemd, Tailscale, Prometheus, Grafana

- Building a private Raspberry Pi 5 dashboard that monitors **system health, Docker services, network checks, and infrastructure diagnostics** from a custom web UI.
- Implemented a **FastAPI backend and Next.js frontend** with health endpoints, action logs, reboot controls, and a Tailscale-gated deployment model.
- Used real hardware and Linux services to practice production-style operations, including secure command execution, service management, and observability.

Personal Website & Engineering Archive

Next.js, TypeScript, Tailwind CSS, content architecture •

- Designed and built a personal website to present **selected engineering work, case studies, technical notes, recipes, and long-form writing**.
- Restructured project presentation to separate polished selected work from smaller experiments and learning projects, improving how public work is positioned.
- Implemented reusable layouts and content organization patterns using TypeScript, Tailwind CSS, and modern frontend tooling.

EDUCATION

University of Maryland, College Park

B.S. Computer Science, Minor in Astronomy

2019 — 2023